

UNIVERSITY OF ENGINEERING AND TECHNOLOGY

Lahore, Pakistan

Department of Computer Engineering

PROJECT PROPOSAL

Inventory Management System

A Python & PyQt5 Desktop Application for Automated Inventory Tracking

Proposed By	Mishkat e Noor
Roll No	2026-CYS(S)-06
Course	Programming Fundamentals Lab
Session	28-06-2026
Submitted To	Hafiz Mubashir Imran
Institution	UET Lahore

1. Introduction

This document serves as a formal project proposal for the development of an Inventory Management System as part of the Programming Fundamentals Lab course at the University of Engineering and Technology, Lahore. The proposal outlines the background of the idea, the gap it addresses, the proposed system, the planned scope of features, tools to be used, and the expected timeline for completion.

The system will be a desktop application built using Python and the PyQt5 graphical framework. It is intended to replace inefficient manual stock-keeping methods with a structured, automated solution that any user can operate without technical expertise.

2. Background & Motivation

Inventory management is a routine operational need across businesses, academic labs, and small organisations. Despite its importance, many setups still rely on paper-based logs or informal spreadsheets to track stock. These methods work at a very small scale but quickly become unreliable as the number of products grows.

The motivation for this project comes from observing how easily errors creep into manual records — a mistyped quantity, a forgotten deletion, or an overwritten entry can lead to stock discrepancies that take hours to resolve. There is a clear opportunity to build a simple yet effective automated system that eliminates these risks.

As a student of the Programming Fundamentals Lab, this project also serves as a practical application of core concepts taught in the course — including functions, classes, file handling, and building user interfaces — in a context that solves a real problem.

3. Proposed System

The proposed system is a single-window desktop application named the Inventory Management System. It will present users with a clean interface showing a live table of all inventory items alongside a form for entering or editing product details.

3.1 Core Idea

The application will allow users to maintain a product inventory by performing the following actions from a single screen:

- Add new products by filling in a name, quantity, and price and clicking a button
- View all existing inventory items in a structured, scrollable table
- Select any item from the table to edit its details and save the update
- Delete items with a confirmation step to prevent accidental removal

- Search through the inventory by typing a product name in a search box
- Have all changes automatically saved so data is never lost on close

3.2 System Modules

Module	Role
main.py	Entry point — launches the Qt application and opens the main window.
inventory_gui.py	GUI layer — manages all user interactions, form state, table rendering, and button events.
inventory_data.py	Data layer — handles reading from and writing to the JSON storage file, and generates unique item IDs.
ui/main_window.ui	Layout file created in Qt Designer defining the window structure and widget placement.
data/inventory.json	Local file where all inventory records are stored and retrieved persistently.

4. Scope of Work

The following table defines what is included in this project and what falls outside its current scope:

In Scope	Out of Scope
Desktop GUI application (Windows/Linux/macOS)	Web or mobile application
Add, view, update, delete inventory items	Barcode or QR code scanning
Real-time name-based search and filtering	Multi-user or networked access
Automatic JSON-based local data persistence	Cloud storage or database backend
Input validation and delete confirmation dialog	PDF or CSV report generation
Status bar feedback after every operation	Stock alert or notification system
Qt Designer .ui layout file	User authentication or login system

5. Tools & Technologies

Tool / Technology	Version	Purpose
Python	3.x	Primary language for all application logic and file handling
PyQt5	5.15.11	GUI framework for building the desktop interface
Qt Designer	Bundled	Visual editor for creating the window layout (.ui file)
JSON (stdlib)	Built-in	Lightweight format for local persistent data storage
PyCharm IDE	Latest	Development environment for writing and running the code

6. Development Plan & Timeline

Week	Activity	Output
Week 1	Requirement Gathering	Finalised list of features and user interaction flows
Week 2	UI Design (Qt Designer)	Completed main_window.ui with all widgets named and placed
Week 3	Core Development	Working Python modules: main.py, inventory_gui.py, inventory_data.py
Week 4	Testing & Bug Fixes	All features verified, edge cases handled, stable application
Week 5	Documentation & Submission	README, project report, and proposal finalised and submitted

7. Expected Deliverables

Upon completion of this project, the following deliverables will be submitted:

- A fully functional Inventory Management System desktop application
- Python source code across three well-structured modules (main.py, inventory_gui.py, inventory_data.py)
- A Qt Designer .ui layout file (ui/main_window.ui) for the main window

- A sample data file (data/inventory.json) demonstrating the storage format
- A README.md file with installation and usage instructions
- This project proposal document
- A separate project report documenting the completed system

8. Potential Challenges

Challenge	Mitigation Plan
Learning PyQt5 signal-slot mechanism	Study official PyQt5 documentation and small example programs first
Linking Qt Designer .ui file to Python code	Use <code>uic.loadUi()</code> at runtime to avoid manual XML parsing
Handling concurrent edits or file corruption	Use try/except around all file operations; create the data directory automatically if missing
Keeping the table in sync with in-memory data	Refresh the table after every add, update, and delete operation

9. Future Enhancements

While the current proposal focuses on a minimal viable product suitable for the course, the following enhancements are planned for future iterations if time and resources permit:

- Barcode and QR code scanning for faster item lookup and stock updates
- Category and supplier management with advanced filtering options
- Low-stock threshold alerts with visual indicators in the table
- Export functionality to generate PDF or CSV inventory reports
- SQLite or cloud database backend for multi-user and multi-device support
- Analytics dashboard showing total stock value and sales trends

10. Approval & Sign-Off

I respectfully submit this proposal for review and approval by the course instructor. I am confident that this project is achievable within the given timeframe and will serve as a strong demonstration of the programming skills developed during the Programming Fundamentals Lab course.

Submitted By:
Mishkat e Noor

Approved By:
Course Instructor

Roll No: 2026-CYS(S)-06

Course: Programming Fundamentals Lab

UET Lahore, 2026

Programming Fundamentals Lab

Department of Cyber Security

UET Lahore

Signature

Signature & Date